ITM SKILLS UNIVERSITY

Department of Computer Science & Engineering

SOFTWARE ENGINEERING & PROJECT MANAGEMENT

PETPAL

Case Study No. 64

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Functional Requirements (FR-01 to FR-12), Non-Functional Requirements (NFR-01 to NFR-08) & Traceability Matrix

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SUBMITTED BY

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Project: PetPal — Appointment Booking and Records App for a Veterinary Clinic Chain
Case Study: No. 64
Subject: Software Engineering & Project Management

SECTION 1

1. Introduction, Purpose & Scope

This Software Requirements Specification (SRS) establishes the formal functional and non-functional baseline for **PetPal**, a web-based veterinary practice management SaaS platform designed for a 5-clinic chain in Bengaluru: *Indiranagar Care Center, Koramangala Veterinary Hospital, Whitefield Specialist Clinic, Jayanagar Wellness Center,* and *Bellandur Outpatient Center*.

> **CRITICAL ARCHITECTURAL BOUNDARY:**
> The Chief Veterinarian has mandated: *"Clinical notes and detailed medical histories are maintained in the existing clinic EMR."* PetPal strictly manages appointment scheduling, pet demographics, and vaccination tracking. **No ClinicalNote entity exists in PetPal.**

SECTION 2

2. Exact Functional Requirements (FR-01 to FR-12)

TABLE 2.1: FUNCTIONAL REQUIREMENTS SPECIFICATION

ID	REQUIREMENT	PROCESSING LOGIC & DESCRIPTION	PRIORITY
FR-01	Registration & Login	The system shall allow pet owners to register and log in securely.	MUST
FR-02	Pet Profile Management	The system shall allow pet owners to create and update pet profiles (Breed, DOB, Weight, Microchip ID).	MUST
FR-03	Vaccination Info Display	The system shall display vaccination information associated with a pet and compute status badges.	SHOULD
FR-04	Clinics & Vets Directory	The system shall allow pet owners to view available clinics (5 Bengaluru locations) and veterinarians.	SHOULD
FR-05	Available Slot Matrix	The system shall display real-time available 30-minute appointment slots.	MUST
FR-06	Appointment Booking	The system shall allow pet owners to book an appointment with a 10-minute temporary checkout hold.	MUST
FR-07	Cancel / Reschedule	The system shall allow pet owners to cancel or reschedule appointments according to clinic rules (≥ 4h prior).	SHOULD
FR-08	Payment Processing	The system shall process online appointment payments through a payment gateway.	MUST
FR-09	Vaccination Reminders	The system shall send vaccination reminders based on recorded vaccination due dates (14d, 3d, 1d prior).	MUST
FR-10	Notifications Dispatch	The system shall send appointment confirmation and reminder notifications via SMS/Email/Push.	MUST
FR-11	Clinic Staff Dashboard	The system shall provide a clinic dashboard for authorized clinic staff to manage queues.	SHOULD

ID	REQUIREMENT	PROCESSING LOGIC & DESCRIPTION	PRIORITY
FR-12	Legacy EMR Exchange	The system shall exchange required appointment/vaccination information with existing clinic software without duplicating clinical notes.	MUST

SECTION 3

3. Non-Functional Requirements (NFR-01 to NFR-08)

TABLE 3.1: NON-FUNCTIONAL REQUIREMENTS SPECIFICATION

NFR ID	CATEGORY	SPECIFICATION STATEMENT	METRIC TARGET
NFR-01	Performance	95% of normal user requests should complete within 3 seconds.	Latency ≤ 3.0s (95th %)
NFR-02	Availability	At least 99% monthly availability excluding planned maintenance.	Monthly uptime ≥ 99.0%
NFR-03	Privacy	Personal and pet-owner data shall only be accessible to authenticated and authorized users.	RBAC claim token audit
NFR-04	Security	Passwords shall use secure one-way hashing (bcrypt / Argon2).	Salted hash code audit
NFR-05	Reminder Reliability	At least 99% of scheduled reminders should be successfully queued by their scheduled time.	Queue delivery logs ≥ 99.0%
NFR-06	Usability	A new pet owner should be able to complete appointment booking within 5 minutes without assistance.	Usability lab time ≤ 300s
NFR-07	Scalability	The system should support the 5-clinic chain and allow additional clinics without redesigning the core system.	Provision 6th branch via config
NFR-08	Recovery	After a recoverable service failure, normal operation should be restored within 30 minutes.	RTO ≤ 30 min, RPO ≤ 15 min

SECTION 4 & 5

4. MoSCoW Prioritization & 5. Traceability Matrix

TABLE 5.1: REQUIREMENTS TRACEABILITY MATRIX (RTM) & TEST MAPPING

REQ ID	REQUIREMENT STATEMENT	MOSCOW	UML REF	UI SCREEN	PLANNED TEST CASE
FR-01	Registration & Login	Must	User Class, UC-01	Screens 01, 02, 18	TC-01, TC-02
FR-02	Pet Profile Management	Must	Pet Class, UC-02	Screens 04, 05, 06	TC-03, TC-04
FR-03	Vaccination Info Display	Should	Vaccination Class, UC-03	Screens 07, 23	TC-05
FR-04	Clinics & Vets Directory	Should	Clinic, Vet, UC-04	Screens 09, 10	TC-06
FR-05	Available Slot Matrix	Must	Appointment Engine, UC-05	Screens 11, 25	TC-07

REQ ID	REQUIREMENT STATEMENT	MOSCOW	UML REF	UI SCREEN	PLANNED TEST CASE
FR-06	Appointment Booking	Must	Appointment, SD-01, AD-01	Screens 08, 12, 14	TC-08
FR-07	Cancel / Reschedule	Should	State Machine SD-02, UC-07	Screens 15, 21	TC-09, TC-10
FR-08	Payment Processing	Must	Payment Class, UC-08	Screen 13	TC-11, TC-12
FR-09	Vaccination Reminders	Must	Vaccination Engine, SD-02	Screen 24, Daemon	TC-13
FR-10	Notifications Dispatch	Must	Notification Class, UC-10	Screen 16	TC-14
FR-11	Clinic Staff Dashboard	Should	Dashboard Service, UC-11	Screens 19, 20, 26	TC-15
FR-12	Legacy Metadata Exchange	Must	IntegrationClient, UC-13	Backend Gateway	TC-16